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Batch 15

Algorithm

Main Program

1. Start.
2. Display the menu:
    - Change date format.
    - Print all substrings.
    - Exit.
3. Read the user's choice.
4. Use a `switch` statement.
5. Call the appropriate function.
6. Repeat the menu until Exit is selected.
7. Stop.

Function 1: Change Date Format

1. Read the date in `dd/mm/yyyy` format.
2. Extract the day, month, and year.
3. Store month abbreviations in an array:
   `Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec`.
4. Check whether the month is between 1 and 12.
5. Convert the four-digit year to two digits using `year % 100`.
6. Display the date as `dd-mmm-yy`.

Example:

```
12/10/2026 → 12-Oct-26
```

Function 2: Print All Substrings

1. Read a string.
2. Find its length.
3. Use an outer loop to select the starting position.
4. Use a second loop to select the ending position.
5. Use an inner loop to print characters from the starting position to the ending position.
6. Repeat until all possible substrings are displayed.

For `abc`:

```
a
```

```
ab
abc
b
bc
c
```

Test Case Table

Choice	Input	Expected Output
1	12/10/2026	12-Oct-26
1	01/01/2025	01-Jan-25
1	25/12/2030	25-Dec-30
1	15/13/2026	Invalid month!
2	abc	a, ab, abc, b, bc, c
2	ab	a, ab, b
2	C	C
3	—	Program terminates
5	—	Invalid choice!

C Program

```
#include <stdio.h>
#include <string.h>

void changeDateFormat(void);
void printSubstrings(void);

int main()
{
    int choice;

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Change date format (dd/mm/yyyy to dd-mmm-
yy)\n");
        printf("2. Print all substrings of a string\n");
        printf("3. Exit\n");
```

```
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                changeDateFormat();
                break;

            case 2:
                printSubstrings();
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice!\n");
        }

    } while (choice != 3);

    return 0;
}

/* Function to change date format */
void changeDateFormat(void)
{
    int day, month, year;

    char months[12][4] = {
        "Jan", "Feb", "Mar", "Apr",
        "May", "Jun", "Jul", "Aug",
        "Sep", "Oct", "Nov", "Dec"
    };

    printf("Enter date in dd/mm/yyyy format: ");
    scanf("%d/%d/%d", &day, &month, &year);
    getchar();

    if (month < 1 || month > 12)
    {
```

```
        printf("Invalid month!\n");
        return;
    }

    printf("Date in dd-mmm-yy format: ");
    printf("%02d-%s-%02d\n",
           day, months[month - 1], year % 100);
}

/* Function to print all substrings */
void printSubstrings(void)
{
    char str[100];
    int i, j, k, len;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    str[strcspn(str, "\n")] = '\0';

    len = strlen(str);

    printf("All substrings are:\n");

    for (i = 0; i < len; i++)
    {
        for (j = i; j < len; j++)
        {
            for (k = i; k <= j; k++)
            {
                printf("%c", str[k]);
            }

            printf("\n");
        }
    }
}
```

```
Last login: Tue Oct 13 12:20:43 on ttys000
bhavya@Bhavyas-MacBook-Air ~ % nano day45.c
bhavya@Bhavyas-MacBook-Air ~ % clang day45.c -o day45
bhavya@Bhavyas-MacBook-Air ~ % ./day45

----- MENU -----
1. Change date format (dd/mm/yyyy to dd-mmm-yy)
2. Print all substrings of a string
3. Exit
Enter your choice: 1
Enter date in dd/mm/yyyy format: 12/10/2026
Date in dd-mmm-yy format: 12-Oct-26

----- MENU -----
1. Change date format (dd/mm/yyyy to dd-mmm-yy)
2. Print all substrings of a string
3. Exit
```

Sample Output 1

```
Enter your choice: 2
Enter a string: abc
All substrings are:
a
ab
abc
b
bc
c

----- MENU -----
1. Change date format (dd/mm/yyyy to dd-mmm-yy)
2. Print all substrings of a string
3. Exit
Enter your choice: 3
Exiting program...
bhavya@Bhavyas-MacBook-Air ~ %
```

Sample output 2